$$\begin{aligned}\mathcal{C}(\tau) &= \sum_i g_i^2 \left\langle \hat{\sigma}_z^{(i)}(\tau)\hat{\sigma}_z^{(i)} \right\rangle_{\hat{\rho}\mathcal{E}} \\ &= \sum_i g_i^2 \,\mathrm{Tr}_{\mathcal{E}} \left\{ \left[\frac{1}{Z} \prod_i \mathrm{e}^{-\hat{H}_{\mathcal{E}}/k_{\mathrm{B}}T} \right] \hat{\sigma}_z^{(i)}(\tau)\hat{\sigma}_z^{(i)} \right\}. \end{aligned} \tag{5.176}$$